

PRE-MEDICAL : LEADER & ACHIEVER COURSE PHASE - MLB, MLK, MLM, MLP, MLQ, MLR, MAZE, MAZF, MAZG, MAZH, MAZM, MAZP, MAZQ, MAZR, MAZU, MAZV, MAAZ, MAAW, SPARK PRO

Test Booklet Code

This Booklet contains 24 pages.

L4

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

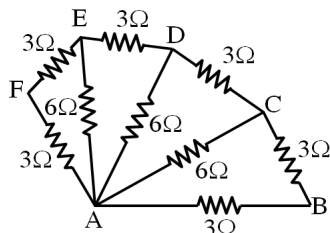
Candidate's Signature : _____ Invigilator's Signature : _____

YOUR TARGET IS TO SECURE GOOD RANK IN PRE-MEDICAL 2025

SUBJECT : PHYSICS

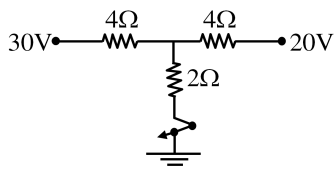
Topic : SYLLABUS-4

1. The equivalent resistance between the points A and B of the network shown in figure is :



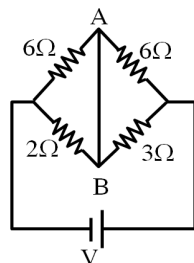
- (1) 3Ω (2) 6Ω (3) 2Ω (4) 1Ω

2. Find out current passing through switch when it is closed ?



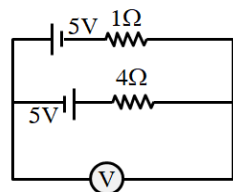
- (1) $\frac{25}{4}$ A (2) $\frac{25}{2}$ A
(3) 25A (4) Zero

3. Find out direction of current in wire AB.



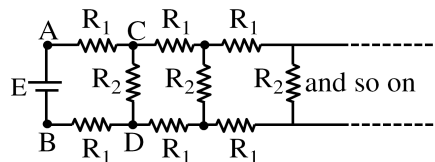
- (1) A to B
(2) B to A
(3) Zero
(4) Depends on EMF of Battery

4. Find out Reading of Ideal Voltmeter.



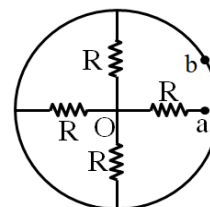
- (1) 5V (2) 3V (3) Zero (4) 1V

5. Find the equivalent resistance between A and B of the network extending off to the infinity shown in the figure for $R_1 = 1\Omega$, $R_2 = 4\Omega$:



- (1) 4Ω (2) 1Ω (3) 2Ω (4) 8Ω

6. The equivalent resistance between points a and b of a network shown in the figure is given by :-

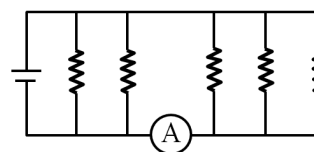


- (1) $\frac{3}{4}R$ (2) $\frac{4}{3}R$ (3) $\frac{5}{4}R$ (4) $\frac{4}{5}R$

7. The length of a given cylindrical wire is increased by 100%. Due to the consequent decrease in diameter the change in the resistance of the wire will be :

- (1) 300% (2) 200% (3) 100% (4) 50%

8. Five identical resistors, each of value 1100Ω are connected to a 220V battery as shown. The reading of the ideal ammeter A is :



- (1) $1/5$ A (2) $2/5$ A (3) $3/5$ A (4) $4/5$ A

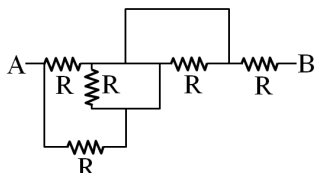
9. A voltmeter has resistance of 2000Ω and it can measure upto 2V. If we want to increase its range to 10V then the required resistance in series will be :

- (1) 2000Ω (2) 9000Ω
(3) 6000Ω (4) 8000Ω

10. A (100 W, 200 V) bulb is connected to a 160 volts supply. The power consumption would be :

(1) 100W (2) 125W
(3) 64W (4) 80W

11. What is the equivalent resistance between A and B?

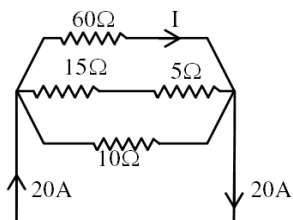


(1) $\frac{3R}{2}$ (2) $2R$ (3) R (4) $\frac{R}{2}$

12. There are 8.4×10^{22} free electrons per cm^3 in copper. The current in the wire is 0.21 A ($e = 1.6 \times 10^{-19}$ C). Then the drift velocity of electrons in a copper wire of 1 mm^2 cross section, will be :

(1) $2.12 \times 10^{-5} \text{ m/s}$
(2) $0.78 \times 10^{-5} \text{ m/s}$
(3) $1.56 \times 10^{-5} \text{ m/s}$
(4) None of these

13. Calculate current (I) in 60Ω :

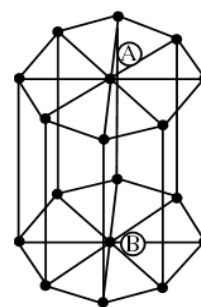


(1) 1A (2) 6A
(3) $\left(\frac{10}{3}\right) \text{ A}$ (4) 2A

14. Two wires of equal diameters of resistivities ρ_1 and ρ_2 and lengths ℓ_1 and ℓ_2 respectively, are joined in series. The equivalent resistivity of the combination is :

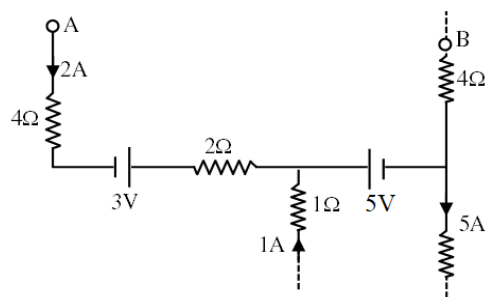
(1) $\frac{\rho_1 \ell_1 + \rho_2 \ell_2}{\ell_1 + \ell_2}$ (2) $\frac{\rho_1 \ell_2 + \rho_2 \ell_1}{\ell_1 - \ell_2}$
(3) $\frac{\rho_1 \ell_2 + \rho_2 \ell_1}{\ell_1 + \ell_2}$ (4) $\frac{\rho_1 \ell_1 - \rho_2 \ell_2}{\ell_1 - \ell_2}$

15. Find $R_{AB} = ?$, If resistance of each side is R :



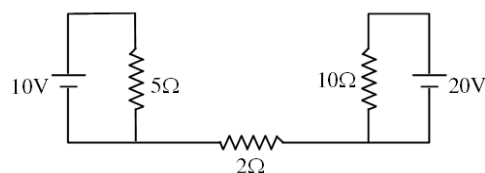
(1) $\frac{3R}{8}$ (2) $\frac{2R}{5}$ (3) $\frac{R}{2}$ (4) $8R$

16. In given network find $V_A - V_B$:-



(1) 6V (2) -6V (3) 5V (4) -10V

17. Find the current through 2Ω :

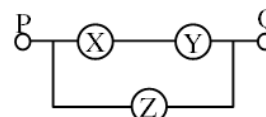


(1) Zero (2) 1 amp (3) 2 amp (4) 4 amp

18. When four identical bulbs of 100 W, 150 V rating are connected in parallel to a 150 V supply the power drawn by them will be :

(1) 25 W (2) 400 W (3) 100 W (4) 50 W

19. Three voltmeters X, Y and Z of resistance R, 2R and 3R respectively are connected as shown in figure. A potential difference is applied across P and Q, then reading of voltmeters are V_X , V_Y and V_Z then :



(1) $V_X = V_Y = V_Z$ (2) $V_X = V_Y \neq V_Z$
(3) $V_X \neq V_Y = V_Z$ (4) $V_X \neq V_Y \neq V_Z$

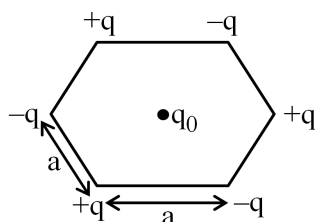
20. A uniform wire of resistance 50Ω is cut into 5 equal parts. These parts are now connected in parallel. The equivalent resistance of combination is :

(1) 2Ω (2) 10Ω
(3) 250Ω (4) 6250Ω

21. Two identical metal spheres A and B have charges of 12 nC and -6 nC respectively. A neutral identical metal sphere C is touched to A, then to B and finally removed. What will be charged on sphere C finally.

(1) 12 nC (2) 6 nC
(3) 3 nC (4) 0

22. Find net electric force feel by charge q_0 placed at centre of regular hexagon.



(1) $k \frac{2qq_0}{a^2}$ (2) $k \frac{qq_0}{a^2}$
(3) zero (4) $\frac{3}{2}k \frac{qq_0}{a^2}$

23. The number of electrons in 2C of charge is :-

(1) 5×10^{29}
(2) 125×10^{17}
(3) 1.6×10^{19}
(4) 9×10^{11}

24. A point charge q is placed at $x = 0$. The electric field vector at $x = L$ is $\vec{E}$. When another charge $4q$ is placed at $x = 2L$, what is the new electric field vector at $x = L$?

(1) $4\vec{E}$ (2) $5\vec{E}$
(3) $-3\vec{E}$ (4) $-2\vec{E}$

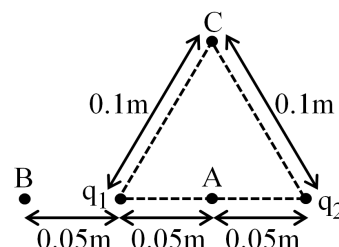
25. For charges $2\text{ }\mu\text{C}$ and $3\text{ }\mu\text{C}$, if force between them for some separation in air is F , then force between them in a medium of permittivity ϵ will be :-

(1) $\frac{\epsilon_0}{\epsilon}F$ (2) $\frac{\epsilon}{\epsilon_0}F$
(3) $\epsilon\epsilon_0F$ (4) $\frac{F}{\epsilon_0\epsilon}$

26. Five balls numbered (1) to (5) are checked pair wise. Pair (3, 2), (2, 4) and (4, 3) show electrostatic attractions while pairs (2, 1) and (4, 5) show repulsion. Ball (1) and ball (3) :-

(1) Ball (1) $\rightarrow$ may be +ve.
Ball (3) $\rightarrow$ must be neutral.
(2) Ball (1) $\rightarrow$ must be (+ve).
Ball (3) $\rightarrow$ may be neutral.
(3) Ball (1) $\rightarrow$ must be (-ve).
Ball (3) $\rightarrow$ may be (+ve).
(4) Ball (1) $\rightarrow$ must be (-ve).
Ball (3) $\rightarrow$ must be (-ve).

27. Two point charges $q_1 = +10^{-8}\text{ C}$ and $q_2 = -10^{-8}\text{ C}$, respectively are placed 0.1 m apart then, ratio of magnitudes of electric fields at A and C is :-



(1) 4 : 1 (2) 1 : 4
(3) 8 : 1 (4) 1 : 8

28. How does the superposition principle apply to the electric potential of charges ?

(1) Potential is scalar and simply add up.
(2) Potential cancel out
(3) Potential depend on the direction.
(4) All of these

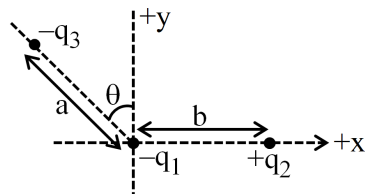
29. If the net electric flux through a closed surface is zero can we conclude that the electric field at every point on the surface is zero ?

- (1) Yes
- (2) No
- (3) Only if the charge inside is zero.
- (4) It depends on the symmetry of the charge distribution.

30. If two bodies are rubbed and one of them acquires q_1 charge and another acquired q_2 charge, then ratio $q_1 : q_2$ is :-

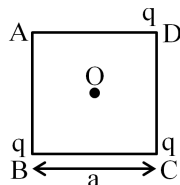
- (1) 1 : 2 (2) 2 : 1 (3) -1 : 1 (4) 1 : 4

31. Three charges are placed as shown find components of force on $-q_1$ are :-



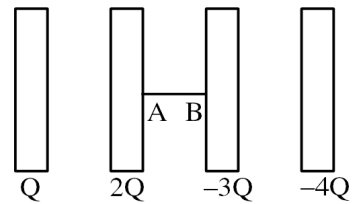
- (1) $k \left(\frac{q_1 q_2}{b^2} + \frac{q_1 q_3 \sin \theta}{a^2} \right) \hat{i}, \frac{k q_1 q_3 \cos \theta}{a^2} (-\hat{j})$
- (2) $\left(k \frac{q_1 q_2}{b^2} \right) \hat{i}, \frac{k q_1 q_3}{b^2} (-\hat{j})$
- (3) $\frac{k q_1 q_3}{a^2} \hat{i}, \frac{k q_2 q_1}{b^2} (-\hat{j})$
- (4) $\frac{k q_1 q_3}{b^2} \hat{i}, \frac{k q_1 q_3}{a^2} \cos \theta (-\hat{j}) + \frac{k q_1 q_2}{b^2}$

32. Equal charges q are placed at the three corners of square ABCD of side "a". Find the ratio of potential at A to centre O is ?



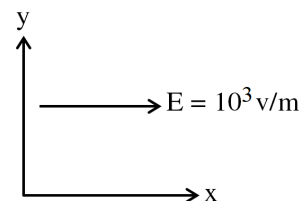
- (1) $\frac{2\sqrt{2}+1}{3\sqrt{2}}$ (2) $\frac{2+\sqrt{2}}{6}$
- (3) $\frac{2\sqrt{2}+1}{6}$ (4) $\frac{2\sqrt{2}+1}{3}$

33. Four large metal plates are shown in fig. The middle two are then connected through a wire find the charge that will flow through the wire.



- (1) $5Q$ from A to B
- (2) $\frac{5Q}{2}$ from A to B
- (3) $5Q$ from B to A
- (4) No charge will flow

34. Find the workdone by an external agent in slowly shifting a charge $q = 1 \mu\text{C}$ in the electric field $\vec{E} = 10^3 \hat{i} \frac{\text{V}}{\text{m}}$ from the point P(1, 2) m to Q(3, 4) m :

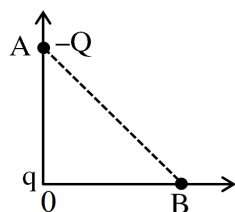


- (1) $2 \times 10^{-3} \text{ J}$
- (2) $4 \times 10^{-3} \text{ J}$
- (3) $-2 \times 10^{-3} \text{ J}$
- (4) $-4 \times 10^{-3} \text{ J}$

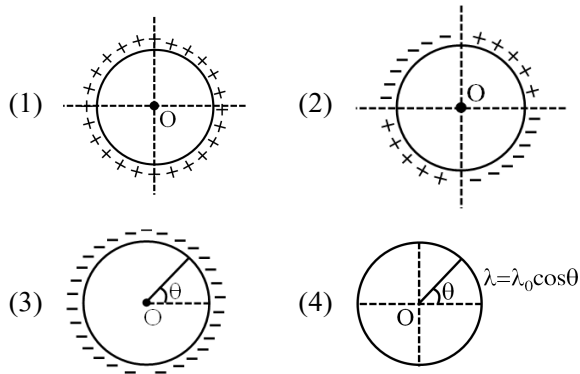
35. The electric potential in a region is given by $V = 3x^2 + 4x$. When an electron is placed at the origin it will experience

- (1) Force along positive x
- (2) Zero
- (3) Force along negative x
- (4) ∞

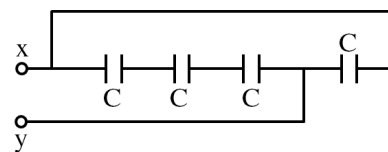
36. The work required to move point charge $-Q$ from the point $A(0, a)$ to another point $B(a, 0)$ along the straight path AB



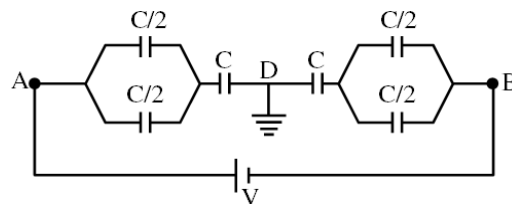
- (1) $\frac{kqQ}{a}$ (2) $\frac{kqQ}{a^2}$
 (3) Zero (4) $-\frac{qQ}{4\pi\epsilon_0\sqrt{2}a}$
37. Electric potential $V = -5x + 3y + \sqrt{15}z$, then find magnitude of the electric field.
 (1) $3\sqrt{2}$ (2) $4\sqrt{2}$ (3) $5\sqrt{2}$ (4) 7
38. There is no net charge at any point A the conductor and any excess charge must reside at the B. There A and B refer to :
 (1) inside, outer surface
 (2) surface, centre
 (3) outer surface, inside
 (4) centre, inside
39. If xJ of work must be done to move electric charge equal to $4C$ from a place where potential is $-10V$ to another place where potential is $5V$, find the value of x .
 (1) 100 J (2) 50 J (3) 60 J (4) 30 J
40. In which of the following figure electric field at point O is non-zero ?



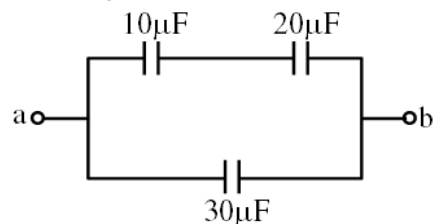
41. Find equivalent capacitance between x and y is -



- (1) $\frac{C}{4}$ (2) $\frac{3C}{4}$ (3) $\frac{C}{3}$ (4) $\frac{4C}{3}$
42. In the circuit as shown, find the potentials at A and B :



- (1) $+V$, Zero (2) $+V$, $-V$
 (3) $+\frac{V}{2}$, $-\frac{V}{2}$ (4) $+\frac{V}{3}$, $-\frac{V}{3}$
43. Let k and x are dielectric constant and dielectric strength of dielectric material respectively. A suitable material for using as a dielectric in a capacitor has :-
 (1) Low ' k ', High ' x ' (2) Low ' k ', Low ' x '
 (3) Low ' x ', High ' k ' (4) High ' k ', High ' x '
44. Potential difference between a and b is 50 volt. Find total charge stored in the circuit :



- (1) $\frac{5500}{3} \mu C$ (2) $750 \mu C$
 (3) $3000 \mu C$ (4) $1280 \mu C$
45. A sphere of radius 1 cm has a potential 8000 V, then energy density near its surface will be :
 (1) $64 \times 10^5 J/m^3$ (2) $8 \times 10^3 J/m^3$
 (3) $32 J/m^3$ (4) $2.83 J/m^3$

SUBJECT : CHEMISTRY

Topic : SYLLABUS-4

46. Which of the following statement is correct ?
- (1) $(\Delta_{\text{sol}}H^\circ)_{\text{CuSO}_4} = (\Delta_{\text{lattice}}H^\circ)_{\text{CuSO}_4} + (\Delta_{\text{Hydration}}H^\circ)_{\text{CuSO}_4}$
 - (2) $(\Delta_{\text{sol}}H^\circ)_{\text{CuSO}_4} = (\Delta_{\text{lattice}}H^\circ)_{\text{CuSO}_4} - (\Delta_{\text{Hydration}}H^\circ)_{\text{CuSO}_4}$
 - (3) $(\Delta_{\text{sol}}H^\circ)_{\text{CuSO}_4} = -(\Delta_{\text{lattice}}H^\circ)_{\text{CuSO}_4} + (\Delta_{\text{Hydration}}H^\circ)_{\text{CuSO}_4}$
 - (4) All of the above
47. Which the following reaction is correct representation of enthalpy of neutralization reaction ?
- (1) $\text{HCl}_{(\text{aq})} + \text{NaOH}_{(\text{aq})} \rightarrow \text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\ell)} ; \Delta H_{\text{neut.}} = -57.1 \text{ kJ eq}^{-1}$
 - (2) $\text{H}^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})} \rightarrow \text{H}_2\text{O}_{(\ell)} ; \Delta H_{\text{neut.}} = -57.1 \text{ kJ eq}^{-1}$
 - (3) $\text{HCl}_{(\text{aq})} + \text{LiOH}_{(\text{aq})} \rightarrow \text{LiCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\ell)} ; \Delta H_{\text{neut.}} = -13.7 \text{ k Cal g eq}^{-1}$
 - (4) All of the above
48. If temperature of a reaction is increased from 10°C to 100°C then how many times rate of reaction will become :-
- (1) 2^2
 - (2) 2^4
 - (3) 2^9
 - (4) 2^6
49. Slope of which plot will give the value of activation energy :-
- (1) K v/s T
 - (2) $\frac{1}{\text{K}} \text{ v/s T}$
 - (3) $\ln \text{K v/s T}$
 - (4) $\ln \text{K v/s } \frac{1}{\text{T}}$
50. The vapour pressure of pure benzene and toluene are 160 and 60 torr respectively. The mole fraction of toluene in vapour phase in contact with equimolar solution of benzene and toluene is :-
- (1) 0.5
 - (2) 0.6
 - (3) 0.27
 - (4) 0.73
51. The color of phenolphthalein indicator in acid solution is
- (1) Pink
 - (2) Yellow
 - (3) Colourless
 - (4) Orange
52. 1L of 1M HCl and 1L of 1M H_2SO_4 are completely neutralized with NaOH separately. The heat liberated in two cases are x and y calories respectively. Then which of the following relation is correct ?
- (1) $x = y$
 - (2) $x > y$
 - (3) $x \ll y$
 - (4) $y = 2x$
53. **Assertion** : Negatively charged As_2S_3 colloidal sol is obtained by double decomposition method using As_2O_3 and H_2S .

$$\text{As}_2\text{O}_3 + 3\text{H}_2\text{S} \rightarrow \text{As}_2\text{S}_3 + 3\text{H}_2\text{O}$$
Reason : Adsorption of As^{3+} ions take place on As_2S_3 when As_2O_3 solution is saturated with H_2S .
- (1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
 - (2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
 - (3) **Assertion** is true but **Reason** is false.
 - (4) Both **Assertion** and **Reason** are false.
54. In one molal solution that contains 0.5 mole of a solute, there is :
- (1) 500 g of solvent
 - (2) 100 mL of solvent
 - (3) 1000 g of solvent
 - (4) 500 mL of solvent
55. In water saturated air the mole fraction of water vapour is 0.02. If the total pressure of the saturated air is 1.2 atm, the partial pressure of dry air is :
- (1) 1.18 atm
 - (2) 1.76 atm
 - (3) 1.176 atm
 - (4) 0.98 atm
56. In an aqueous solution of NaCl mole fraction of solvent is 0.9 then relative lowering in vapour pressure will be –
- (1) 0.2 atm
 - (2) 0.1
 - (3) 0.2
 - (4) 0.05

57. What will be the molarity of 49% w/w H_2SO_4 solution having density = 1.2 g/mL.

- (1) 3 M (2) 6 M
(3) 1 M (4) 12 M

58. Which of the following aqueous solution has highest osmotic pressure at 25°C

- (1) 0.1 M AlCl_3 (2) 0.01 M $\text{Al}_2(\text{SO}_4)_3$
(3) 0.5 M Glucose (4) 0.20 M CaCl_2

59. A solution of two volatile liquids form maximum boiling azeotrope at a particular composition, then select the incorrect statement :-

- (1) This solution shows negative deviation from Raoult's law
(2) HNO_3 and H_2O may form this type of solution.
(3) Vapour pressure of solution at azeotrope composition is maximum
(4) At this composition, components cannot be separated by fractional distillation.

60. What will be the degree of dissociation of 0.1 M $\text{Mg}(\text{NO}_3)_2$ solution if Van't Hoff factor is 2.74?

- (1) 75% (2) 87% (3) 100% (4) 92%

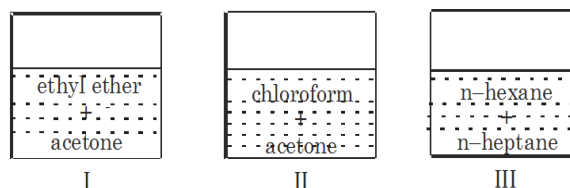
61. Which of the following solvent has maximum value of cryoscopic constant?

- (1) Water (H_2O)
(2) Cyclohexane (C_6H_{12})
(3) Carbon disulphide (CS_2)
(4) Carbon tetrachloride (CCl_4)

62. Bromoform has a normal freezing point of 7.9°C and its $K_f = 14.4^\circ\text{C/molal}$. A solution of 1.3 g of an unknown in 100 g of bromoform freezes at 5.3°C . What is the molecular weight of the unknown :

- (1) 7.2 (2) 72 (3) 144 (4) 14.4

63. State T(true)/F(false) for following statements regarding given liquid solutions :



(a) Final volume of solutions if equal volume of compounds are mixed, is in the order:

$$V_I = V_{II} < V_{III}$$

(b) ΔS_{mix} for solution I and solution II is positive but for solution III it is zero.

(c) Enthalpy change of mixing is negative for solution II and solution III but positive for solution I.

- (1) TFF (2) FFF
(3) TTF (4) FTT

64. Ratio of solubilities of gases N_2 & O_2 in water (as mole of gas in large volume of water) from air at 25° & 1 atm will be, if air is 20 % by mole of O_2 & 80% by mole of N_2

Given : $K_H(\text{N}_2) = 2 \times 10^4 \text{ atm}$

$K_H(\text{O}_2) = 10^4 \text{ atm}$

- (1) 8 : 1 (2) 1 : 8
(3) 2 : 1 (4) 1 : 2

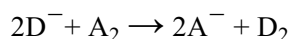
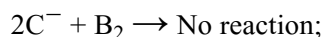
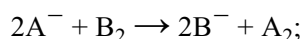
65. The electrolytic decomposition of dilute sulphuric acid with platinum electrode, cathodic reaction is:-

- (1) Reduction of $\text{H}_2\text{O} / \text{H}^+$
(2) Oxidation of SO_4^{2-}
(3) Reduction of SO_4^{2-}
(4) Oxidation of H_2O

66. In an experiment, 0.04 F was passed through 400 mL of a 1M solution of NaCl . What would be the pH of the solution after the electrolysis :-

- (1) 8 (2) 10 (3) 13 (4) 6

67. The following facts are available;



Which of the following order is correct :-

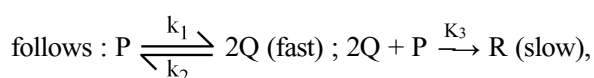
$$(1) E_{C^-|C_2}^0 > E_{B^-|B_2}^0 > E_{A^-|A_2}^0 > E_{D^-|D_2}^0$$

$$(2) E_{C^-|C_2}^0 < E_{B^-|B_2}^0 < E_{A^-|A_2}^0 < E_{D^-|D_2}^0$$

$$(3) E_{C^-|C_2}^0 < E_{B^-|B_2}^0 > E_{A^-|A_2}^0 > E_{D^-|D_2}^0$$

$$(4) E_{C^-|C_2}^0 > E_{B^-|B_2}^0 < E_{A^-|A_2}^0 < E_{D^-|D_2}^0$$

68. The reaction mechanism for the reaction $P \rightarrow R$ is as



the rate law for the main reaction ($P \rightarrow R$) is :

$$(1) k_1[P][Q] \quad (2) k_1k_2[P]$$

$$(3) \frac{k_1k_3[P]^2}{k_2} \quad (4) k_1k_2[P]^2$$

69. $SO_2Cl_2 \rightarrow SO_2 + Cl_2$ is a first order gaseous reaction with $K = 2.3 \times 10^{-5} \text{ sec}^{-1}$ at 320°C . The % of SO_2Cl_2 decomposed on heating for 90 minute is : ($10^{0.054} = 1.13$)

$$(1) 11.5 \quad (2) 1.15$$

$$(3) 0.118 \quad (4) 18.11$$

70. A reaction takes place in three steps with individual rate constant and activation energy,

	Rate constant	Activation energy
Step 1	k_1	$E_{a1} = 180 \text{ kJ/mol}$
Step 2	k_2	$E_{a2} = 80 \text{ kJ/mol}$
Step 3	k_3	$E_{a3} = 50 \text{ kJ/mol}$

$$\text{overall rate constant, } k = \left(\frac{k_1k_2}{k_3} \right)^{2/3}$$

overall activation energy of the reaction will be :

$$(1) 140 \text{ kJ/mol} \quad (2) 150 \text{ kJ/mol}$$

$$(3) 130 \text{ kJ/mol} \quad (4) 120 \text{ kJ/mol}$$

71. The initial rate of reaction $A + 5B + 6C \rightarrow 3L + 3M$ has been determined by measuring the rate of disappearance of A under the following conditions :-

Experiment No.	$[A]_0 (M)$	$[B]_0 (M)$	$[C]_0 (M)$	Initial rate ($M \text{ min}^{-1}$)
1.	0.02	0.02	0.02	2.08×10^{-3}
2.	0.01	0.02	0.02	1.04×10^{-3}
3.	0.02	0.04	0.02	4.16×10^{-3}
4.	0.02	0.02	0.04	8.32×10^{-3}

Determine the rate law ?

$$(1) \text{ Rate} = k[A][B][C]^2$$

$$(2) \text{ Rate} = k[A][C]^2$$

$$(3) \text{ Rate} = k[A][B][C]$$

$$(4) \text{ Rate} = k[A][B][C]^{1/2}$$

72. Two different species A and B show first order disintegration having the half lives 0.231 and 0.3465 second, respectively. The ratio of initial concentration of A and B is 3 : 2. The value of C_A/C_B after three half lives of A will be :-

$$(1) 3/4$$

$$(2) 2/3$$

$$(3) 4/5$$

$$(4) 1/7$$

73. Calculate value of emf of cell :-



$$\left[\text{Given } \frac{2.303RT}{F} = 0.06 \right]$$

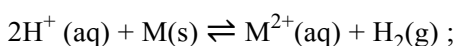
$$(1) 0.010 \text{ V}$$

$$(2) 0.20 \text{ V}$$

$$(3) 0.09 \text{ V}$$

$$(4) 0.45 \text{ V}$$

74. Find SRP of M^{+2}/M electrode from following information :-



$$\Delta G^0 = +96.5 \text{ KJ at } 298 \text{ K}$$

$$(1) 0.5 \text{ V}$$

$$(2) -0.5 \text{ V}$$

$$(3) 1 \text{ V}$$

$$(4) -1 \text{ V}$$

75. Based on the following information arrange four metals A, B, C and D in order of decreasing ability to act as reducing agents :
- (I) Only A, B and C react with 1 M HCl to give $H_2(g)$
- (II) When C is added to solutions of the other metal ions, metallic B and D are formed
- (III) Metal C does not reduce A^{n+}
- (1) $C > A > B > D$
- (2) $C > A > D > B$
- (3) $A > C > D > B$
- (4) $A > C > B > D$
76. For saturated solution of Ag_2CrO_4 at infinite dilution $(\lambda_m^\infty)_{Ag^+} = 127 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and $(\lambda_m^\infty)_{CrO_4^{2-}} = 273 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively and specific conductance of solution is $2 \times 10^{-2} \text{ ohm}^{-1} \text{ cm}^{-1}$; calculate solubility product of Ag_2CrO_4 :-
- (1) 0.05 (2) 1.5
- (3) 0.037 (4) 2.2×10^{-4}
77. Quantity of charge required to reduce 12.3 gm of Nitrobenzene to aniline is (molar mass of nitrobenzene = 123):-
- (1) 57900 C (2) 28950 C
- (3) 19300 C (4) 9650 C
78. $PbO_2 \rightarrow PbO \quad \Delta G_{298K} < 0$
 $SnO_2 \rightarrow SnO \quad \Delta G_{298K} > 0$
 Most probable oxidation state of Pb and Sn will be :
- (1) Pb^{4+}, Sn^{4+} (2) Pb^{4+}, Sn^{2+}
- (3) Pb^{2+}, Sn^{2+} (4) Pb^{2+}, Sn^{4+}
79. Electrode potential will be less for hydrogen electrode at pH.
- (1) 2 (2) 3
- (3) 4 (4) 5
80. **Statement I** :- During electrolysis of aqueous sodium acetate solution, the molar ratio of gases formed at cathode and anode is 1 : 3.
- Statement II** :- Acetate ion discharges at anode and H^+ ion, at cathode.
- (1) Both **Statement I** and **Statement II** are true.
- (2) **Statement I** is true but **Statement II** is false.
- (3) **Statement I** is false but **Statement II** is true.
- (4) Both **Statement I** and **Statement II** are false.
81. **Statement I** :- In the Daniel cell, if concentrations of Cu^{+2} and Zn^{+2} ions are doubled, the E.M.F. of cell does not change.
- Statement II** :- Active mass of solids and pure liquids does not effect the E.M.F. of cell.
- (1) Both **Statement I** and **Statement II** are true but **Statement II** is NOT the correct explanation of **Statement I**
- (2) **Statement I** is true but **Statement II** is false.
- (3) **Statement I** is false but **Statement II** is true.
- (4) Both **Statement I** and **Statement II** are true and **Statement II** is the correct explanation of **Statement I**
82. What will happen if a aluminium spoon is used to stir a solution of copper nitrate ?
- (1) The Spoon will get coated with copper
- (2) Alloy of copper and aluminium is formed
- (3) The solution becomes blue
- (4) Spoon will disappear after some time
83. The equation that is incorrect is :-
- (1) $(\Lambda_m^0)_{H_2O} = (\Lambda_m^0)_{HCl} + (\Lambda_m^0)_{NaOH} + (\Lambda_m^0)_{NaCl}$
- (2) $(\Lambda_m^0)_{NaBr} - (\Lambda_m^0)_{NaCl} = (\Lambda_m^0)_{KBr} - (\Lambda_m^0)_{KCl}$
- (3) $(\Lambda_m^0)_{KCl} - (\Lambda_m^0)_{NaCl} = (\Lambda_m^0)_{KBr} - (\Lambda_m^0)_{NaBr}$
- (4) All are incorrect

84. Match the following

1	Neutralization Titration	P	formation of coordination compound
2	Precipitation titration	Q	oxidation reduction reaction
3	Redox titration	R	formation of salt and water
4	Complexometric titration	S	formation of precipitate

(1) 1-R, 2-S, 3-Q, 4-P (2) 1-Q, 2-R, 3-P, 4-S

(3) 1-R, 2-P, 3-S, 4-Q (4) 1-S, 2-Q, 3-R, 4-P

85. Match the half-life in column-I with the order in column-II for the reaction $A \rightarrow \text{product}$.

Column-I		Column-II	
(A)	$t_{1/2} \propto a$	(P)	First order
(B)	$t_{1/2} \propto \frac{1}{a}$	(Q)	III rd order
(C)	$t_{1/2} \propto \frac{1}{a^2}$	(R)	II nd order
(D)	$t_{1/2} = \text{constant}$	(S)	Zero order

(1) A-P, B-R, C-Q, D-S

(2) A-S, B-R, C-Q, D-P

(3) A-S, B-R, C-P, D-Q

(4) A-S, B-Q, C-R, D-P

86. **Statement I** :- The temperature coefficient of a single-step reaction may be negative.

Statement II :- The rate of reaction having negative order with respect to a reactant decreases with the increase in concentration of that reactant.

(1) **Statement I** is incorrect but **Statement II** is correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) Both **Statement I** and **Statement II** are correct.

87. The rate constant is given by the equation $K = P A e^{-E_a/RT}$. Which factor should register a decrease for the reaction to proceed more rapidly.

(1) P

(2) A

(3) E_a

(4) T

88. Freshly prepared precipitate sometimes gets converted to colloidal solution by :-

(1) Coagulation

(2) Peptisation

(3) Electrolysis

(4) Diffusion

89. **Assertion** :- Aquatic species are more comfortable in warm water than in cold water.

Reason :- K_H values for both N_2 and O_2 increases with increase in temperature.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

90. Compare boiling point of given solutions.

(I) $10^{-4}M$ NaCl (II) $10^{-3}M$ NaCl

(III) $10^{-2}M$ NaCl (IV) $10^{-4}M$ Urea

(1) III > I > II > IV

(2) I > II > III > IV

(3) III > II > I > IV

(4) II > I > III > IV

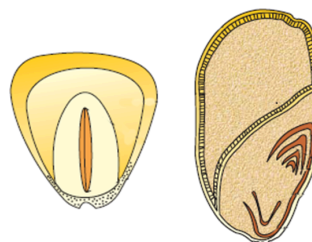
SUBJECT : BIOLOGY

Topic : SYLLABUS-4

91. The arrangement of flowers on the floral axis is termed as –
- (1) Venation
 - (2) Phyllotaxy
 - (3) Inflorescence
 - (4) Placentation
92. In which of following seed coat is membranous and generally fused with the fruit wall :-
- (1) Gram
 - (2) Castor
 - (3) Coconut
 - (4) Maize
93. Scutellum is found in seeds of :-
- (1) Maize
 - (2) Gram
 - (3) Pea
 - (4) Bean
94. In seeds of cereals the outer covering of endosperm separates the embryo by which layer–
- (1) Epicarp
 - (2) Fibrous layer
 - (3) Aleurone layer
 - (4) Testa
95. Given below are the two statements :
- Statement-I** : In monocot plants primary root is short lived.
- Statement-II** : In monocot like wheat fibrous root system is present.
- (1) Both statements are correct
 - (2) Only statement -I is correct
 - (3) Only statement -II is correct
 - (4) Both statements are incorrect

96. **Assertion (A)** : In majority of dicotyledonous plants, direct elongation of radicle leads to the formation of primary root.
- Reason (R)** : In monocotyledonous plants, fibrous root system is present.

- (1) Both A and R are true and R is correct explanation of A.
 - (2) A is true but R is false.
 - (3) A is false but R is true.
 - (4) Both A and R are true and R is not correct explanation of A.
97. Which of the following statement is correct ?
- (1) In mango and coconut, fruits, developed from monocarpellary inferior ovaries.
 - (2) In coconut fruit, the epicarp is fibrous.
 - (3) In castor seeds, endosperm is a food storing structure
 - (4) Aleurone layer is present in seeds of gram.
98. Select the correct option with respect to given diagrams



- (1) Seed coat is membranous
- (2) Seed coat fused with fruit wall
- (3) Plumule and radicle are not enclosed in sheaths.
- (4) Both (1) and (2)

99. Find out incorrect match :-

- (1) Vexillary aestivation in petal - Bean
- (2) Epitpalous stamens - Brinjal
- (3) Apocarpous gynoecium - Lotus
- (4) Basal Placentation - Marigold

100. Which of the following statement is not true ?

- (1) Polyadelphous stamens are found in *Citrus*.
- (2) Twisted aestivation is found in petals of lady finger flower.
- (3) In cymose inflorescence the flowers are borne in acropetal order.
- (4) False septum is found in ovary of mustard flower.

101. Match the column-I with column-II and select the correct option.

Column-I		Column-II	
(A)	$\oplus \text{ } \text{K}_{2+2} \text{C}_4 \text{A}_{2+4} \text{G}_{(2)}$	I.	<i>Petunia</i>
(B)	$\% \text{ } \text{K}_{(5)} \text{C}_{1+2+(2)} \text{A}_{(9)+1} \text{G}_1$	II.	Lupin
(C)	$\oplus \text{ } \text{K}_{(5)} \text{C}_{(5)} \text{A}_5 \text{G}_{(2)}$	III.	Tulip
(D)	$\text{Br } \oplus \text{ } \text{P}_{(3+3)} \text{A}_{3+3} \text{G}_{(3)}$	IV.	Mustard

- (1) A-I, B-II, C-IV, D-III
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-I, C-II, D-III
- (4) A-IV, B-II, C-I, D-III

102. In __A__ type of phyllotaxy, a single leaf arises at each node, as in china rose, mustard and __B__ plants.

- (1) [A] Alternate, [B] *Calotropis*
- (2) [A] Opposite, [B] Guava
- (3) [A] Alternate, [B] Sunflower
- (4) [A] Opposite, [B] *Alstonia*

103. Fruit is developed without fertilisation is called :-

- (1) Apospory
- (2) Parthenogenesis
- (3) Parthenocarp
- (4) Polyembryony

104. In which plant part two to four vascular bundles with radial and exarch xylem present :

- (1) Monocot stem
- (2) Dicot stem
- (3) Monocot root
- (4) Dicot root

105. On the basis of following terms find out the type of leaf :

Bulliform cells, mesophyll undifferentiated, Parallel venation.

- (1) Isobilateral leaf
- (2) Only dorsiventral leaf
- (3) Both isobilateral and dorsiventral leaf
- (4) Only dicotyledonous leaf

106. The cortex of dicotyledonous roots is consisted of :-

- (1) Parenchyma
- (2) Sclerenchyma
- (3) Collenchyma
- (4) (2) and (3) both

107. Stomatal apparatus include :-

- (1) Stomatal aperture
- (2) Guard cell
- (3) Subsidiary cell
- (4) All of these

108. In dicot root many cells of epiblema protude in the form of :-

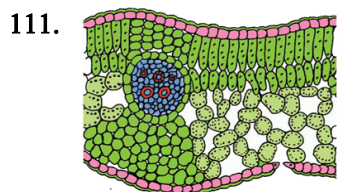
- (1) Cortex
- (2) Multicellular root hair
- (3) Unicellular root hair
- (4) Hypodermis

109. In dicot stem the hypodermis is mainly consist of :

- (1) Parenchyma
- (2) Collenchyma
- (3) Sclerenchyma
- (4) Xylem

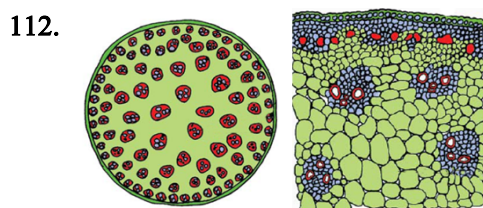
110. What type of vascular bundles are found in monocotyledons stem ?

- (1) Open vascular bundles
- (2) Conjoint closed vascular bundles
- (3) Radial vascular bundles
- (4) Conjoint open vascular bundles



Which of following option is incorrect about given diagram :

- (1) Loosely arranged mesophyll is known as spongy parenchyma.
- (2) Xylem are present towards abaxial surface.
- (3) Dorsiventral leaf.
- (4) More stomata towards abaxial surface.



Which of the following is not correct about diagram ?

- (1) Hypodermis is sclerenchymatous
- (2) Vascular bundles are conjoint closed
- (3) Exarch protoxylem
- (4) Pith is indistinct

113.

Identify the given diagram and select correct option about diagram.

- (1) Conjoint open vascular bundle
- (2) Present in stem of sun flower
- (3) Unable to form secondary xylem and phloem tissue
- (4) Present in monocot roots

114. Read carefully the statements about root and identify the incorrect statement :

- (1) The anatomy of the monocot root is similar to the dicot root in many respect.
- (2) Monocot roots generally show secondary growth.
- (3) In dicot root thick walled parenchymatous cells are called pericycle.
- (4) Casparian strips are present in endodermis.

115. Which of the following statement are not incorrect:-

- (a) Epidermis is outermost layer always made up of multilayered parenchymatous cells
 - (b) Shoot hair are generally multicellular and branched structure
 - (c) Root hair are only present at the tip of root
 - (d) Subsidiary cell in stomata are modified endodermal cell
- (1) a & b
 - (2) a, b, c
 - (3) b & d
 - (4) b, c, d

116. What is the name of the oldest viable seed excavated from Arctic Tundra ?

- (1) *Strigms*
- (2) *Lupinus arcticus*
- (3) *Phoenix dectyfera*
- (4) *Orobanche*

117. In some cereals such as rice and wheat, pollen grains lose viability within (a) of their release and in some members of Rosaceae, Leguminosae and Solanaceae, they maintain viability for (b) .

- (1) (a) 30 minutes, (b) months
- (2) (a) 30 minutes, (b) years
- (3) (a) 30 seconds, (b) months
- (4) (a) months, (b) 30 minutes

118. In Autogamy :

- (1) Plant should be dioecious
- (2) Bisexual flowers requires synchrony in pollen release and stigma receptivity
- (3) Anther and stigma should not lie close to each other
- (4) Plant should be monoecious with unisexual flowers

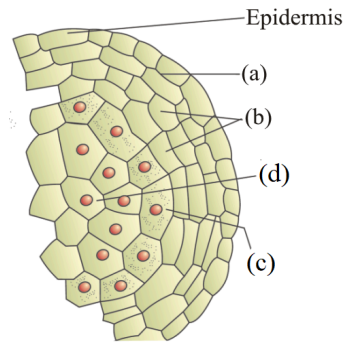
119. The pollen tube releases both male gametes inside the cytoplasm of which degenerating cell of embryo sac ?

- (1) Antipodal cell
- (2) Synergid cell
- (3) Egg cell
- (4) Central cell

120. In *Zostera* pollination takes place through _____

- (1) Wind
- (2) Water
- (3) Animals
- (4) Insects

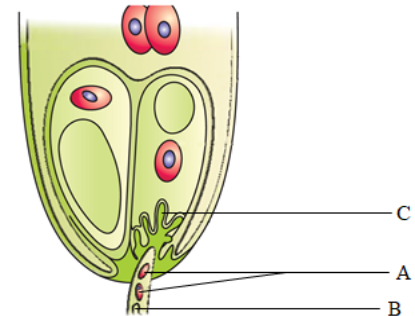
121.



Identify (a), (b), (c) and (d) in above diagram

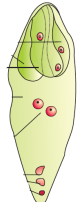
- (1) (a) endothecium, (b) tapetum, (c) middle layers, (d) microspore mother cells
- (2) (a) microspore mother cells, (b) middle layers, (c) tapetum, (d) endothecium
- (3) (a) Endothecium, (b) middle layers, (c) tapetum, (d) microspore mother cells
- (4) (a) tapetum, (b) endothecium, (c) microspore mother cells, (d) middle layers

122. Observe the given diagram carefully and select the correct option in which labelled part A, B and C are correctly identified?



- (1) A-Male gametes
B-Vegetative nucleus
C-Filiform apparatus
- (2) A-Vegetative nucleus
B-Male gametes
C-Filiform apparatus
- (3) A-Synergid
B-Central cell
C-Egg nucleus
- (4) A-Male gametes
B-Central cell
C-Synergid

123. Examine the figure A-D given below and select the right option in which all the four figures A, B, C and D are identified correctly.

	A	B	C	D
1				
2	Pollen grain tetrad	Ovule	mature embryo sac	mature pollen grain
3	Megaspore tetrad	Ovule	mature embryo sac	mature pollen grain
4	Megaspore tetrad	Ovule	mature male gametophyte	mature pollen grain

124. Match the column-I with column-II and select correct option.

Column-I		Column-II	
A.	Funicle	I.	Mass of cells with abundant reserve food materials.
B.	Hilum	II.	Basal part of ovule.
C.	Integument	III.	One or two protective layers of ovule.
D.	Chalaza	IV.	Region where body of ovule fuses with funicle.
E.	Nucellus	V.	Stalk of ovule.

- (1) A-I, B-II, C-III, D-IV, E-V
 (2) A-V, B-IV, C-III, D-II, E-I
 (3) A-IV, B-II, C-I, D-III, E-V
 (4) A-I, B-III, C-V, D-II, E-IV

125. Functionally cross pollination but genetically self pollination is?

- (1) Xenogamy
 (2) Geitonogamy
 (3) Autogamy
 (4) Both (1) and (2)

126. Majority of insect pollinated flowers are :

- (1) Large
 (2) Colourful
 (3) Fragrant and rich in nectar
 (4) All of the above

127. The outermost layer of endosperms of maize grain is called

- (1) Epidermis (2) Tunica
 (3) Aleurone (4) Pericarp

128. In which type of pollination flowers have single ovule in ovary and numerous flowers packed into inflorescence ?

- (1) Wind pollination (2) Water pollination
 (3) Bee pollination (4) Bird pollination

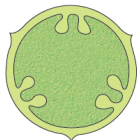
129. Primary endosperm nucleus (PEN) is formed by the fusion of :-

- (1) 2 polar nuclei + 1 synergid cell nucleus
 (2) 1 polar nuclei + 1 antipodal cell nucleus + 1 synergid cell nucleus
 (3) 2 polar nuclei + 1 male gamete nucleus
 (4) 2 antipodal cell nuclei + 1 male gamete nucleus

130. Filiform apparatus are seen in :-

- (1) Egg cell (2) Antipodal cells
 (3) Synergid cells (4) Endothecium cells

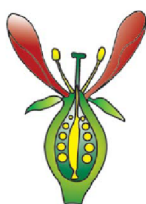
131.



This type of placentation is related to which of the following plant -

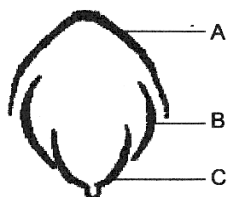
- (1) Pea (2) Mustard
(3) Dianthus (4) Chinrose

132. The following diagram represents the position of the ovary in.



- (1) Guava (2) Cucumber
(3) Plum (4) 1 and 2 both

133. Name the petals A, B and C in vexillary aestivation shown in the below figure.



- (1) A-Standards, B-Wing, C-Perianth
(2) A-Standard, B-Keel, C-Wing
(3) A-Wing, B-Keel, C-Wing
(4) A-Standard, B-Wing, C-Keel

134. **Assertion:** In Cymose inflorescence, Growth of main axis is limited.

Reason: The main axis terminates in a flower.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

135. Read following two statements(I and II) and choose correct option -

Statement - I : Leaves develops at the node and bears a bud in its axil.

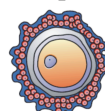
Statement - II : Leaves originate from shoot apical meristems and are arranged in basipetal order.

- (1) Statement I and statement II both are correct
(2) Statement I and statement II both are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

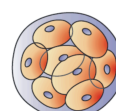
136. Which statement is incorrect regarding female external genitalia ?

- (1) Mons pubis is a cushion of fatty tissue covered by skin and pubic hairs
(2) Labia minora are paired fold of tissue under labia majora
(3) Clitoris lies at the upper junction of labia minora below the urethral opening
(4) Clitoris is a tiny finger like structure.

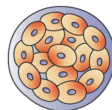
137. The figure below show four stage (a, b, c, d) of human development. Select the option given correct identification together with number of cells present in it :-



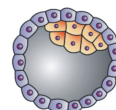
(a)



(b)



(c)



(d)

		Developmental stage	Number of cells
(1)	(a)	Zygote	2-4
(2)	(b)	Blastula	8-16
(3)	(c)	Morulla	16-32
(4)	(d)	Blastocyst	16-32

138. Assertion : The testes of human are situated outside the body in an extra-abdominal pouch called scrotum.

Reason : Scrotum helps in maintaining the low temperature of testes by $2-2.5^{\circ}\text{C}$ less than the normal body temperature, which is necessary for spermatogenesis.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (2) **Assertion** is true but **Reason** is false.
- (3) **Assertion** is false but **Reason** is true.
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

139. Assertion : Absence of menstruation is not a confirmatory indicator of pregnancy.

Reason : Absence of menstruation can happen due to poor health or stress.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (2) **Assertion** is true but **Reason** is false.
- (3) **Assertion** is false but **Reason** is true.
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

140. The first movements of the foetus usually observed during the :

- (1) First trimester of pregnancy
- (2) Fourth month of pregnancy
- (3) Fifth week of pregnancy
- (4) Fifth month of pregnancy

141. How many days, pregnancy last in dogs, elephants and cats ?

- (1) 30 days, 290 days, 60 days
- (2) 30 days, 640 days, 60 days
- (3) 60 days, 640 days, 65 days
- (4) 65 days, 270 days, 30 days

142. Secondary follicle in ovary has-

- (1) Primary oocyte
- (2) Secondary oocyte
- (3) Ootid
- (4) Antrum

143. In oogenesis process, division of primordial germ cells and formation of oogonia occurs at the time of :-

- (1) Before puberty
- (2) Embryo stage
- (3) After puberty
- (4) In adult stage

144. In human beings longest phase of spermatogenesis is :-

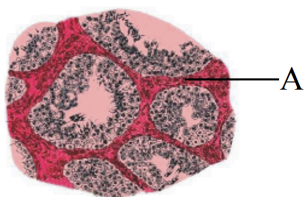
- (1) Resting phase
- (2) Multiplication phase
- (3) Growth phase
- (4) Maturation phase

145. Menstrual cycle is controlled by :-

- (a) Estrogen and progesterone of ovary
 - (b) FSH of anterior pituitary gland
 - (c) LH of adenohypophysis
 - (d) Oxytocin of neurohypophysis
- (1) Only a and b are correct.
 - (2) Only b and d are correct.
 - (3) Only a and c are correct.
 - (4) a, b and c are correct.

- 146.** Which of the following is released by spermatozoa that acts on the corona radiata cells during fertilisation ?
- (1) Relaxin
 - (2) Gonadotropin
 - (3) Hyaluronidase
 - (4) Trypsinogen
- 147.** Level of estrogen and progesterone are minimum at:-
- (1) Follicular phase
 - (2) Ovulation
 - (3) Secretory phase
 - (4) Bleeding phase
- 148.** Choose the incorrect for luteal phase :-
- (1) Transformation of graafian follicle in corpus luteum
 - (2) Large amount of progesterone
 - (3) Maintenance of the endometrium by progesterone
 - (4) Heavy menstruation
- 149.** Which hormone maintain corpus luteum during pregnancy?
- (1) LH
 - (2) FSH
 - (3) hCG
 - (4) Both (1) & (2)
- 150.** In the human female, menstruation can be deferred by the administration of :-
- (1) FSH only
 - (2) LH only
 - (3) Combination of FSH and LH
 - (4) Combination of estrogen and progesterone
- 151.** Common duct formed by union of vas deferens and duct from seminal vesicle is
- (1) urethra
 - (2) tunica-vasculosa
 - (3) ejaculatory duct
 - (4) spermatic duct
- 152.** The mitotic division starts as the zygote moves through the _____ of the oviduct called cleavage towards the uterus.
- (1) Ampulla
 - (2) Isthmus
 - (3) Fimbriae
 - (4) Infundibulum
- 153.** Several mammary ducts join to form a wider A which is connected to B through which milk is sucked-out. Identify the A & B.
- (1) A-Mammary tubule,
B-Mammary ampulla
 - (2) A-Mammary Ampulla,
B-Lactiferous duct
 - (3) A-Lactiferous duct,
B-Mammary ampulla
 - (4) A-Mammary Ampulla,
B-Nipple
- 154.** How many meiosis required to form 160 zygotes?
- (1) 220
 - (2) 320
 - (3) 160
 - (4) 200

155. The given diagram is a cross section of seminiferous tubules. What is the function of structure A?



- (1) Formation of sperms
(2) Nourishment of sperms
(3) Secretion of testosterone
(4) All of the above
156. Which structure does not carry sperms ?
- (1) Vas deferens
(2) Rete testis
(3) Epididymis
(4) Prostate gland
157. Read the following statements (A-D) and choose the correct option.
- (A) Hymen is often torn during the first coitus.
(B) The clitoris is a tiny finger like structure which lies at the lower junction of two labia minora.
(C) Myometrium exhibits strong contraction during delivery of baby.
(D) Several mammary ducts join to form a wider mammary lobe.
- (1) Only (A) and (C) are correct.
(2) (A), (B) and (C) are correct.
(3) (B), (C) and (D) are correct.
(4) (B) and (C) are incorrect.

158. **Statement-I** :- Labia majora extend down from mons pubis & surrounds vaginal & urethra opening.

Statement-II :- Clitoris is finger like projection at upper junction of labia minora above urethral opening.

- (1) Statement I & II both are correct
(2) Statement I & II both are incorrect
(3) Statement I is correct but II is incorrect
(4) Statement I is incorrect but II is correct
159. First haploid cell during spermatogenesis is :-
- (1) Spermatogonia
(2) Primary spermatocyte
(3) Secondary spermatocyte
(4) Spermatids
160. Semen is :-
- (1) Seminal plasma along with sperm
(2) Secretion of seminal vesicle along with sperm
(3) Secretion of epididymis along with sperm
(4) Secretion of androgen along with sperm
161. Primary oocyte complete its first meiotic division in which structure ?
- (1) Primary follicle (2) Corpus luteum
(3) Secondary follicle (4) Tertiary follicle
162. Theca layer is organised into an inner theca interna and outer theca externa in :-
- (1) Primary follicle (2) Secondary follicle
(3) Tertiary follicle (4) Corpus luteum
163. Which hormone shows significant increase at the age of puberty -
- (1) GnRH (2) Growth Hormone
(3) Melanin (4) All of the above

- 164.** The second meiotic division during oogenesis completes:
- (1) Before birth
 - (2) Just after birth
 - (3) At the time of puberty
 - (4) During fertilization
- 165.** Which of the following is a copper releasing IUD ?
- (1) Lippes loop
 - (2) Multiload 375
 - (3) Progestasert
 - (4) LNG-20
- 166.** The medical termination of pregnancy (Amendment) act was enacted in :-
- (1) 1951
 - (2) 1971
 - (3) 2017
 - (4) 2000
- 167.** Test tube baby programme also known :-
- (1) Artificial insemination
 - (2) In vitro fertilisation
 - (3) In vivo fertilisation
 - (4) Gamete intra fallopian transfer
- 168.** Find the odd one out about Saheli
- (1) Contains steroidal preparation
 - (2) Once a week pill
 - (3) Few side effects
 - (4) High contraceptive value
- 169.** __A__ programme create awareness among people about various reproductive related aspects and provides facilities for build up a reproductive healthy society.
- (1) Family planning programme
 - (2) Reproductive and child health care (RCH)
 - (3) Contraceptive methods
 - (4) Lactational amenorrhea
- 170.** Which of the following a natural/traditional contraceptive method ?
- (1) Condoms
 - (2) Coitus interruptus
 - (3) Vaults
 - (4) Diaphragms
- 171.** Emergency contraceptives are effective if used within :-
- (1) 72 hrs of coitus
 - (2) 72 hrs of ovulation
 - (3) 72 hrs of menstruation
 - (4) 72 hrs of implantation
- 172.** Given below some Assisted Reproductive Technologies (ART). In which find out In vitro Fertilisation :-
- (a) ZIFT
 - (b) IUT
 - (c) ICSI
 - (d) AI
 - (e) GIFT
 - (f) IUI
- (1) a, b, c
 - (2) b, c, d
 - (3) d, e, f
 - (4) b, e, f
- 173.** In which of the technique fertilisation occur inside the body.
- (1) ZIFT
 - (2) IUT
 - (3) ICSI
 - (4) A.I.
- 174.** Embryo with up to 8 blastomeres formed due to in vitro fertilization is transferred into -
- (1) Cervix
 - (2) Uterus
 - (3) Fimbriae
 - (4) Fallopian tube
- 175.** Choose the incorrect match pair :-
- | | | |
|-----|---------------|--|
| (1) | Multiload 375 | Suppress sperm motility |
| (2) | Progestasert | Make uterus unsuitable for implantation. |
| (3) | Lippes loop | Make cervix hostile to the sperm |
| (4) | CuT | Suppress sperm motility |

176. Find out the incorrect statement :

- (1) Government of India legalized MTP in 1971.
- (2) A majority of the MTPs are performed legally by unqualified quacks.
- (3) In India, often the female is blamed for the couple being childless.
- (4) In IVF, fertilisation occurs outside the body.

177. **Assertion :** Copper releasing IUD prevent fertilization.

Reason : Copper ions inhibit motility of sperms.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

178. Which of the following statement is incorrect about cervical caps & diaphragms?

- (1) Only to be used by females only.
- (2) Prevents user from contracting STDs
- (3) Applied on cervix part
- (4) A type of barrier method for contraception.

179. **Statement-I :** India was amongst the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal.

Statement-II : Oral administration of small doses of either progestogens or progestogen-estrogen combination is another contraceptive method used by the female.

- (1) Both Statement I and Statement II are correct.
- (2) Statement I is correct and statement II is incorrect.
- (3) Statement I is incorrect and statement II is correct.
- (4) Both Statement I and Statement II are incorrect.

180. Which of the following method of birth control can cause excessive menstrual bleedings pain and inflammation of uterine wall as side effects ?

- (1) Condom
- (2) Cervical cap
- (3) Oral contraceptive
- (4) Intra uterine device (IUDs)



CALL teleMANAS

Toll Free No.

☎ 14416, 1800-8914416

ALLEN De-Stress No.

☎ 0744-2757677 📞 +91-8306998982

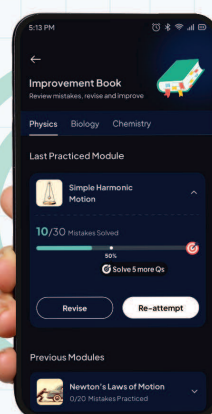
SPACE FOR ROUGH WORK

ALLEN**Turn mistakes into marks**

Track & fix them all in one place with
Improvement Book on the ALLEN app!



SCAN TO
GET AHEAD



PHASE - MLB, MLK, MLM, MLP, MLQ, MLR, MAZE, MAZF, MAZG, MAZH, MAZM, MAZP, MAZQ, MAZR, MAZU, MAZV, MAAZ, MAAW, SPARK PRO

25-03-2025

1016CMD303032240049

ALLEN® CAREER INSTITUTE Pvt. Ltd.

Registered & Corporate Office : 'SANKALP', CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

Ph. : +91-744-3556677, +91-744-2757575 | E-mail : info@allen.in | Website : www.allen.ac.in